Q1.

The diagram below shows types of waves within the electromagnetic spectrum.

Some of the types of waves are represented by letters.

P	**microwaves**	**Q**	**visible light**	**R**	**S**	**gamma rays**

(a) Which letter shows the position of ultraviolet (UV) radiation within the electromagnetic spectrum?

Tick **one** box.

P ☐ **Q** ☐ **R** ☐ **S** ☐

(1)

(b) A special lamp can produce UV radiation.

Which **two** statements describe the electromagnetic waves emitted by a UV lamp?

Tick **two** boxes.

They have a higher frequency than X-rays. ☐

They have the same wave speed as visible light. ☐

They have a longer wavelength than microwaves. ☐

They have a lower frequency than gamma rays. ☐

They have a greater wave speed than radio waves. ☐

(2)

(c) UV radiation is used to treat a vitamin D deficiency.

People should **not** use a UV lamp for long periods of time.

State **two** risks of exposure to high levels of UV radiation.

1. _____

_____

2. _____

_____

(2)

(d) Ionising radiation is used for some medical imaging.

Name **two** types of electromagnetic waves that are used.

1. _____

2. _____

(2)

(Total 7 marks)

Q1.

(a) R

1

(b) they have the same wave speed as visible light

1

they have a lower frequency than gamma rays

1

(c) any **two** from:

- ageing of the skin

  allow sunburn
  burn unqualified is insufficient

- skin cancer

  cancer is insufficient
  allow mutation of genes in skin cells

- eye damage

  allow blindness

2

(d) any **two** from:

- gamma (rays)
- X-rays
- ultraviolet

2

[7]